

Attack Analysis Lab Report

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Introduction

This report displays a comprehensive forensic analysis of a suspected intrusion on a Windows-based host identified as “Maybe Hacked Box.” The organization’s Cyber First Responder collected several forensic artifacts immediately after the incident were declared; an application event log, a security event log, and a packet capture file (PCAP). Based on standard attack-analysis methodology and industry best practices. This investigation focuses on determining the sequences of malicious activities, identifying how the attacker interacted with the system, and proposing actionable remediation steps to prevent recurrence.

The goal of this analysis is to find each forensic source in depth, network traffic and correlate findings across logs, techniques, and identify the attacker’s likely tactics, and procedures (TTPs). Some are emphasized from evidence showing the credential exposure, the command execution activity, and the transfer of a malicious Windows executable over the network. All the findings indicate that the compromised host experienced both unauthorized access and post-exploitation activity consistent with remote control and lateral movement attempts.

This report follows the structure expected of a formal cyber incident analysis: As tools and methodology, application log analysis, security log analysis, network traffic analysis, attack timeline reconstruction, remediation recommendations, and conclusion. The screenshots will be referenced throughout the report are attached in the appendix.

Tools and Methodology

This investigation is employed by three primary forensic sources: Windows event logs (.evt), a Wireshark packet capture (.pcapng), and investigator interpretation through correlation. The event logs were opened and reviewed using the Windows Event Viewer tool, which provided structured access to security events and system generated. These logs allow investigators to privilege-related activity, process creation anomalies, identify service failures, system-level warnings and authentication attempts. All the events were reviewed for unexpected source behavior, relevance to known attack vectors, and suspicious patterns.

Network traffic analysis was performed using Wireshark, a standard packet inspection tool. Analysis included filtering for suspicious keywords such as “password” and “cmd,” reviewing unusual protocol activity, and examining full TCP streams to identify malicious payloads. More specifically, the attention was paid to connections between internal hosts on the unusual ports, repeated payload sizes, and the presence of executable file signatures.

The layered methodology was as follows:

1. Initial Log Review – Identifying high-risk events IDs and anomalies.
2. Network Triage – Search for clear indicators of infiltration.
3. Payload Confirmation – Validate whether binary data or executable content was transferred.

4. Correlation – Connect log irregularities to network events to reconstruct a unified attack sequence.
5. Interpretation – Determine attacker intent and capability based on observed TTPs.

The approach the system went with ensured that the findings are evidence-driven.

Application Log Analysis

The application log over saw several critical insights into the host's behavior leading up to and during the suspected compromise. These purposes of these logs is to serve as a essential layer of visibility into the software components and system services affected during intrusion.

WinMgmt Warnings (Event ID 63)

A entry in the application log appears as a WinMgmt warning (Screenshot 4). It indicates that:

“A provider, CmdTriggerConsumer, has been registered in the Windows Management Instrumentation namespace Root\cimv2 to use the LocalSystem account”

This message is very significant from a forensic standpoint. Registration of a WMI provider under LocalSystem privileges is a technique that is well known. WMI consumers or providers are often modified by attackers to execute commands automatically when specific triggers occur, such as system startup or event patterns. This allows attackers to maintain control without having to rely on traditional startup folders, making things harder to detect.

The reference to CmdTriggerConsumer is interesting. The name itself strongly suggests triggering behavior command based. Combined with network evidence later in the report, it supports the likelihood that the attacker had both system-level access and the ability to execute arbitrary commands.

WinMgmt Warning (Event ID 5603)

Another significant entry (Screenshot 5) indicates:

“The description for Event ID 5603 from source WinMgmt cannot be found.”

Something like this will only happen when system components have been removed, corrupted, or replaced. In the context given it suggests that a confirmed intrusion, missing or invalid log descriptions may indicate tampering. Attackers can either delete or overwrite provider files to conceal activity, break logging functionality, or disable monitoring systems.

Even though they were taken together, the WinMgmt warnings suggest that the attacker interacted with Windows Management Instrumentation. This is likely part of the persistence or the lateral movement preparation. When combined with network evidence of command execution, this makes a large indication of compromise beyond a superficial probing.

Securing Log Analysis

The security event log provides critical information into authentication and system-level anomalies. Several of the captured entries align closely with attacker behavior, particularly during the exploitation and post exploitation stages.

Event ID 4226 – TCP Connection Limit Reached

Security Log Screenshot 1 shows Event ID 4226:

“TCP/IP has reached the security limit imposed on the number of concurrent TCP connect attempts.”

This warning is strongly related to scanning activity, malware behavior, or automated connection attempts. It would take a legitimate ton of user activity to generate enough rapid outbound connections to trigger this limit. Attackers, especially post exploitation tools, frequently open large numbers of outbound TCP sessions for:

- Reconnaissance scans
- Malware propagation
- Command and control beaoning
- Brute force attempts

The timing of this event is connected to the suspicious network packets suggesting that the host may have been trying to communicate multiple times with a remote command server or initiate multiple connections as part of malware functionality.

Event ID 7001 – DHCP and NetBIOS Service Failures

Security Log Screenshot 2 shows Event ID 7001:

“The DHCP Client service depends on the NetBIOS over Tcpip service which failed to start because of the following error: A device attached to the system is not functioning.”

Service failures commonly occurred when system files are unstable, or the network components become unstable. In compromised environments, attackers could disable or corrupt network services to interfere with defensive mechanisms or to make the system use different network paths for a more suitable exploitation.

While this event itself is not conclusively malicious, its presence among the other suspicious indicators and around similar timestamps suggest that a possible attacker had interfere at the network service layer.

Security Log Interpretation

The security events indicate that the service instability possibly induced by an unauthorized modification. Elevated connections activity are consistent with malware or remote control. Although no explicit failed login events were observed in the provided screenshots, the

availability logs still show a pattern of anomalous behavior fully consistent with compromised system.

Network Traffic Analysis (Wireshark)

The PCAP file had to deal with the most definitive evidence of malicious activity. Three distinct findings points conclusively to attacker presence, command execution, credential exposure and malware transfer.

Password Content in TCP Packets

Filtering for the keyword “password” revealed multiple packets (Screenshot 2) exchanged between internal hosts 192.168.134.129 and 192.168.134.132. The term “password” appears in the data payloads suggests plaintext credential transmission. This is a critical security failure. Attackers monitoring the traffic or compromising either endpoint could harvest the credentials directly.

Command-Related Traffic (“cmd”)

Filtering for “cmd” gave two packets (Screenshot 1) of the same large size. Since the presence of the string “cmd” is priority because attackers often use different ways to interact with compromised hosts and one of them is to use command interpreters. Some common tools are reverse shells, PowerShell payloads, or Meterpreter sessions create patterns where commands or partial binary content includes recognizable strings.

Malicious Executable Transferred Over the Network (MZ Header)

The most compelling evidence appears in the TCP stream output (Screenshot 3). Viewing the full stream reveals the unmistakable header of a Windows Portable Executable. “MZ ... This program cannot be run in DOS mode.”

The presence of an MZ header in network traffic indicates that an actual executable file was transmitted over the network. Almost certainly malware or a post-exploitation payload. The size of the stream is persistent with a packed or very light executable.

Attack Timeline Reconstruction

Analysis of the event logs and packet capture reveals a clear sequence of attacker activity. The earliest meaningful indicators appear in the application log, where WinMgmt events show that a WMI consumer was registered under the account. This suggests that the attacker established persistence early by going on and modifying WMI components. Around the sametime as the security log records an Event ID 4226 alerting the system reached the limit of the concurrent TCP connection attempts.

Shortly afterward, the packet shows what suspicious network activity is going on. The packets containing the term “password” suggests the plaintext credential exposure, while packets containing “cmd” suggest remote command over a TCP channel. A TCP stream containing an executable payload with the “MZ” header, proving that a Windows PE file was transferred across the network. The transfer of a binary payload strongly indicates post-exploitation activity, such as malware deployment or preparation for lateral movement. Together, these events show a coordinated attack flow: abnormal network behavior, credential exposure, remote command execution, and the final malware delivery.

Remediation Recommendation

To contain and recover from this compromise, the affected system should be immediately isolated from the network to prevent further command and control activity, because an unauthorized executable was transferred and possibly executed, the safest corrective action is a complete system rebuild. All credentials associated with this host should be reset, and authentication policies should be updated to prevent plain text password exposure.

The unauthorized WMI consumer should be removed, and the WMI repository should be repaired or reset to eliminate any persistence mechanisms. Network monitoring rules should be updated to detect executable transfers, unusual TCP connection spikes, and command patterns in packet payloads. Finally, the organization should implement stronger segmentation, enforce encrypted protocols for authentication, and increase log monitoring to detect similar attacks more spread out in the future.

Conclusion

The combined evidence from the application log, security log, and network capture clearly shows that the host was compromised and used for activity they didn’t authorize. WMI modifications indicate persistence, abnormal TCP behavior suggests automated or malicious communication, and the packet capture reveals plaintext credential exposure, command execution, and the transfer of a Windows executable. This is confirming the active attacker control. Although, the full extent of the compromise cannot be determined from the artifacts alone, the findings are good enough to conclude that the system experienced a multistage intrusion requiring immediate containment, long term security improvement, and remediation.

Appendix

frame contains "cmd"

No.	Time	Source	Destination	Protocol	Length	Info
572	8.121525	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=571349 Ack=1 Win=14600 Len=1460
617	8.121610	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=629749 Ack=1 Win=14600 Len=1460

Frame 572: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{AA48AEC9-CEF5-43EC-B3F1-25D52CAE7321}, id 0

Ethernet II, Src: VMware_3f:28:31 (00:0c:29:3f:28:31), Dst: VMware_7d:f3:78 (00:0c:29:7d:f3:78)

Internet Protocol Version 4, Src: 192.168.134.129, Dst: 192.168.134.132

Transmission Control Protocol, Src Port: 29922, Dst Port: 1241, Seq: 571349, Ack: 1, Len: 1460

Data (1460 bytes)

- Screenshot 1

frame contains "password"

No.	Time	Source	Destination	Protocol	Length	Info
516	8.120586	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=585649 Ack=1 Win=14600 Len=1460
582	8.121529	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=585949 Ack=1 Win=14600 Len=1460
585	8.121530	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=590329 Ack=1 Win=14600 Len=1460

Frame 585: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{AA48AEC9-CEF5-43EC-B3F1-25D52CAE7321}, id 0

Ethernet II, Src: VMware_3f:28:31 (00:0c:29:3f:28:31), Dst: VMware_7d:f3:78 (00:0c:29:7d:f3:78)

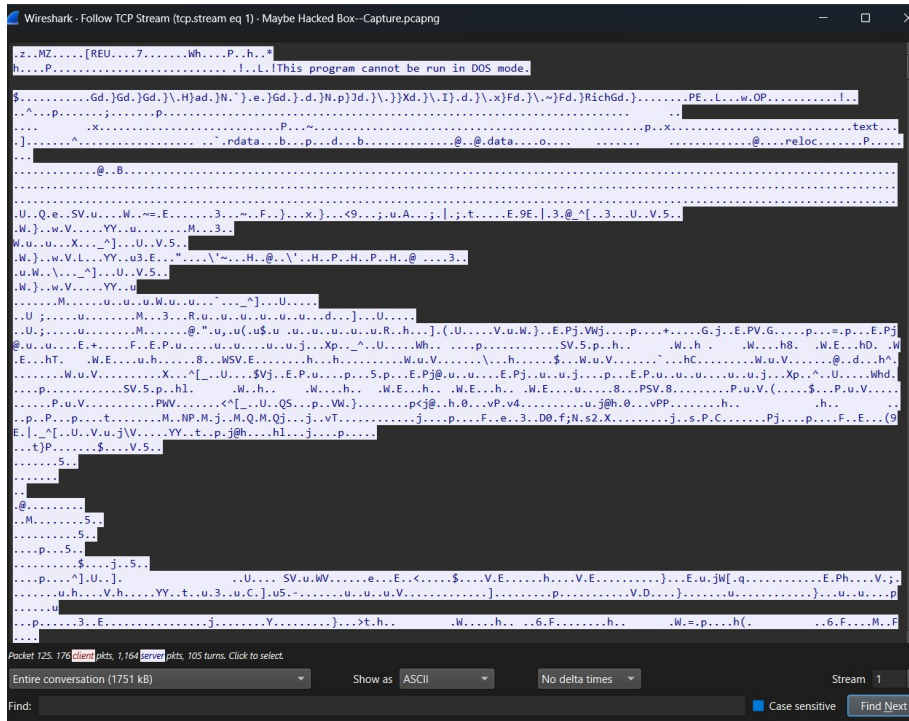
Internet Protocol Version 4, Src: 192.168.134.129, Dst: 192.168.134.132

Transmission Control Protocol, Src Port: 29922, Dst Port: 1241, Seq: 590329, Ack: 1, Len: 1460

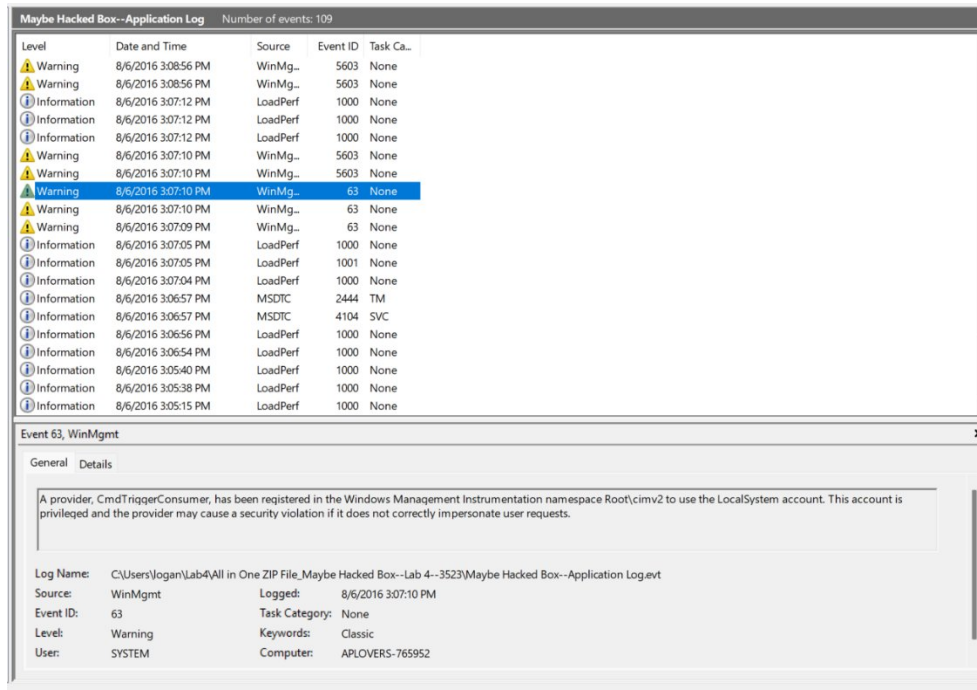
Data (1460 bytes)

- Screenshot 2

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-Screenshot 3



- Screenshot 4

Maybe Hacked Box--Application Log Number of events: 109

Level	Date and Time	Source	Event ID	Task Ca...
Information	8/5/2016 3:13:40 PM	VMTools	105	None
Warning	8/5/2016 3:13:15 PM	WinMg...	5603	None
Warning	8/5/2016 3:13:15 PM	WinMg...	5603	None
Warning	8/5/2016 3:13:15 PM	WinMg...	5603	None
Information	8/5/2016 3:12:47 PM	crypt32	1	None
Information	8/5/2016 3:12:47 PM	crypt32	4	None
Information	8/5/2016 3:12:47 PM	crypt32	2	None
Information	8/5/2016 3:12:47 PM	crypt32	7	None
Information	8/5/2016 3:12:44 PM	MsiInst...	11707	None
Information	8/5/2016 3:12:37 PM	MsiInst...	11707	None
Information	8/5/2016 3:12:36 PM	ESENT	102	General
Information	8/5/2016 3:12:36 PM	ESENT	100	General
Information	8/5/2016 3:11:52 PM	SecurIt...	1800	None
Information	8/5/2016 3:09:22 PM	SecCli	1500	None
Information	8/5/2016 3:09:09 PM	Wmdm...	100	None
Information	8/5/2016 3:08:58 PM	COM+	4156	(113)
Information	8/5/2016 3:08:58 PM	COM+	4156	(113)
Information	8/5/2016 3:08:58 PM	COM+	4156	(113)
Information	8/5/2016 3:08:58 PM	COM+	4156	(113)
Warning	8/5/2016 3:08:56 PM	WinMg...	5603	None

Event 5603, WinMgmt

General Details

The description for Event ID 5603 from source WinMgmt cannot be found. Either the component that raises this event is not installed on your local computer or the installation is corrupted. You can install or repair the component on the local computer.

If the event originated on another computer, the display information had to be saved with the event.

Log Name: C:\Users\jogan\Lab4\All in One ZIP File_Maybe Hacked Box--Lab 4--3523\Maybe Hacked Box--Application Log.evtx
Source: WinMgmt Logged: 8/5/2016 3:13:15 PM
Event ID: 5603 Task Category: None
Level: Warning Keywords: Classic
User: SYSTEM Computer: APLOVERS-765952

- Screenshot 5

Maybe Hacked Box--Security Log Number of events: 191

Level	Date and Time	Source	Event ID	Task Category
Warning	7/8/2017 1:56:46 PM	Tcpip	4226	None
Information	7/8/2017 12:38:43 PM	Service Control Manager	7035	None
Information	7/8/2017 12:38:43 PM	Service Control Manager	7036	None
Information	7/8/2017 12:37:57 PM	Service Control Manager	7036	None
Information	7/8/2017 12:37:52 PM	Service Control Manager	7036	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7036	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7035	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7035	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7036	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7036	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7035	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7035	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7036	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7035	None
Information	7/8/2017 12:37:51 PM	Service Control Manager	7035	None
Information	7/8/2017 12:37:23 PM	Tcpip	4201	None
Information	7/8/2017 12:37:00 PM	vmci	3	None
Information	7/8/2017 12:37:22 PM	EventLog	6005	None
Information	7/8/2017 12:37:22 PM	EventLog	6009	None
Information	8/22/2016 11:32:09 AM	EventLog	6006	None
Information	8/22/2016 11:32:02 AM	Service Control Manager	7036	None

Event 4226, Tcpip

General Details

TCP/IP has reached the security limit imposed on the number of concurrent TCP connect attempts.

Log Name: C:\Users\jogan\Lab4\All in One ZIP File_Maybe Hacked Box--Lab 4--3523\Maybe Hacked Box--Security Log.evtx
Source: Tcpip Logged: 7/8/2017 1:56:46 PM
Event ID: 4226 Task Category: None
Level: Warning Keywords: Classic
User: N/A Computer: APLOVERS-765952

- Screenshot 6

Maybe Hacked Box--Security Log Number of events: 191				
Level	Date and Time	Source	Event ID	Task Category
Error	8/8/2016 4:53:19 PM	Service Control Manager	7001	None
Error	8/8/2016 4:53:19 PM	Service Control Manager	7001	None
Error	8/8/2016 4:53:19 PM	Service Control Manager	7001	None
Error	8/8/2016 4:53:19 PM	Service Control Manager	7001	None
Warning	8/8/2016 4:51:51 PM	vmci	2	None
Warning	8/8/2016 4:51:28 PM	vmci	2	None
Warning	8/8/2016 4:51:21 PM	vmci	2	None
Warning	8/8/2016 4:51:21 PM	vmci	2	None
Information	8/8/2016 4:51:35 PM	EventLog	6005	None
Information	8/8/2016 4:51:35 PM	EventLog	6009	None
Information	8/8/2016 4:49:51 PM	EventLog	6005	None
Information	8/8/2016 4:49:51 PM	EventLog	6009	None
Information	8/8/2016 4:49:18 PM	EventLog	6006	None
Warning	8/7/2016 4:54:14 AM	W32Time	36	None
Information	8/6/2016 3:15:32 PM	Service Control Manager	7036	None
Information	8/6/2016 3:15:32 PM	Service Control Manager	7035	None
Information	8/6/2016 3:15:03 PM	Service Control Manager	7036	None
Information	8/6/2016 3:15:03 PM	Service Control Manager	7036	None
Information	8/6/2016 3:15:03 PM	Service Control Manager	7035	None
Information	8/6/2016 3:15:02 PM	Service Control Manager	7036	None

Event 7001, Service Control Manager	
General	Details
The DHCP Client service depends on the NetBios over Tcpip service which failed to start because of the following error: A device attached to the system is not functioning.	
Log Name:	C:\Users\jogan\Lab4\All in One ZIP File_Maybe Hacked Box--Lab 4--3523\Maybe Hacked Box--Security Log.evtx
Source:	Service Control Manager
Event ID:	7001
Level:	Error
User:	N/A
Task Category:	None
Keywords:	Classic
Computer:	APLOVERS-765952
Logged:	8/8/2016 4:53:19 PM

-Screenshot 7

References

- Microsoft. *Windows Event ID Documentation and Security Auditing Reference*.
<https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/event-id-reference>
- Microsoft. *Windows Management Instrumentation (WMI) Architecture and Event Logging*.
<https://learn.microsoft.com/en-us/windows/win32/wmisdk/wmi-start-page>
- Wireshark Foundation. *Wireshark User Guide and Protocol Reference*.
<https://www.wireshark.org/docs/>
- NIST. *NIST Special Publication 800-61 Revision 2: Computer Security Incident Handling Guide*.
<https://csrc.nist.gov/publications/detail/sp/800-61/rev-2/final>
- MITRE Corporation. *MITRE ATT&CK® Framework: Techniques T1047 (WMI), T1059 (Command Execution), T1105 (Exfiltration)*.
<https://attack.mitre.org/>
- Volatility Foundation. *Digital Forensics Reference for Memory and Malware Behavior*.
<https://www.volatilityfoundation.org/>

